بسم الله الرحمن الرحيم

Ibn Sina University

Faculty of Medicine

First-year- semester one

Introduction to Medical Education

TBL 1: 07.12.2025

Problem-Based Learning PBL

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❖ Intended Learning outcomes

- **By the end of this session, each student will be able to:**

1. Define Problem-Based Learning.
2. Describe the different steps and phases of PBL.
3. Explain the role and responsibilities of group members during the PBL.
4. Identify the advantages and disadvantages of Problem-Based Learning.

Outlines of the sessions

1. Preparatory phase: advance assignment about problem-based learning
2. Readiness assurance test with IF AT
   I. Individual readiness assurance test iRAT.
   II. Team readiness assurance test tRAT.
3. Appealing.
4. Clarifying lecture.
5. Team application tAPP with IF AT.

Individual Readiness Assurance Test iRAT

iRAT (1)

❖ **In a PBL course, students are presented with a complex case involving multiple symptoms. What is the role of the facilitator during this process?**

a) To provide all the answers directly.

b) To guide discussions and help students explore their ideas.

c) To evaluate each student's individual performance only.

d) To take over the discussion if it becomes unproductive.

iRAT (2)

- A group of medical students is presented with a case study of a patient exhibiting symptoms of diabetes. What is the first step they should take in a Problem-Based Learning session?

a) Jump directly to treatment options

b) Identify what they already know about diabetes.

c) Review textbooks for answers.

d) Discuss the case with their instructor.

iRAT (3)

❖ **Which of the following best describes the PBL process?**

a) Presenting information followed by testing.

b) Identifying a problem, researching, and proposing solutions.

c) Completing tasks in isolation.

d) Focusing on theoretical knowledge only.

iRAT (4)

What is one of the primary advantages of Problem-Based Learning (PBL)?

a) It focuses solely on the memorization of facts.

b) It promotes long-term knowledge retention and understanding.

c) It eliminates the need for teamwork.

d) It requires less time than traditional learning methods.

iRAT (5)

❖ During the first PBL session for a case of "a 45-year-old with jaundice," the group spent time clarifying terms like "scleral icterus" and "acholic stools." They then brainstormed possible mechanisms from infection to blockage. According to the 7-Jump model, what are the next TWO critical steps they must perform before adjourning for self-study?

a) Assign research topics to each member and schedule the next meeting.

b) Systematically analyze and categorize their brainstorming ideas to generate a problem statement, and then define their learning objectives (learning issues).

c) Immediately research the most likely diagnosis online and present findings to the group.

d) Propose a definitive treatment plan and role-play delivering the diagnosis to the patient.

Team readiness assurance test tRAT.

The same questions as the Individual Readiness Assurance Test iRAT

Appealing

Clarifying lecture

❖ Definition

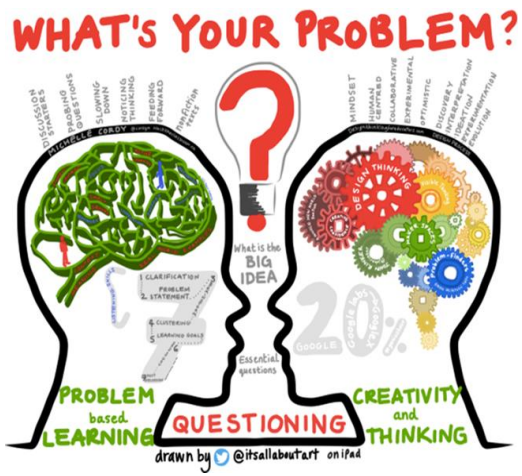
Problem-based learning: Presentation of Problem (Real-world case scenario) as the stimulus for learning, enabling students to understand the relevance of underlying science knowledge and principles in practice.

❖ Major criteria

- ✓ Student-centered.
- ✓ Small group teaching.
- ✓ Active learning methods.
- ✓ Case-based.

Phases and steps of PBL

- **Step 1:** **Identify** and **clarify** unfamiliar terms presented in the scenario
  - ✓The scribe **lists** those that remain unexplained after discussion
- **Step 2:** **Define** the problem(s) to be discussed; students may have different views on the issues, but all should be considered
  - ✓The scribe **records** a list of agreed-upon problems
- **Step 3:** "**Brainstorming**" session: to **discuss** the problem(s), **suggesting** possible explanations based on prior knowledge; students draw on each other's knowledge and identify areas of incomplete knowledge
  - ✓The scribe **records** all discussions.



- **Step 4: Structure and hypothesis** by reviewing steps 2 and 3, and arrange explanations into provisional solutions, and systematically analyze and categorize their brainstorming ideas.
- ✓The scribe **organizes** the explanations and restructures.
- **Step 5: Organize** learning objectives.
- ✓The tutor ensures that learning objectives are **focused**, **achievable**, **comprehensive**, and **appropriate.**

Step 6: Private study (all students gather information related to each learning objective).

Step 7: Group **shares and evaluates the results** of private study (students identify their learning resources and share their results).

- The tutor **checks** the learning process and **assesses** the group.

❖ Roles and responsibilities of PBL group members

I. Tutor/ facilitator:

1. Facilitates and guides learning.
2. Creates an optimum learning environment.
3. Listens carefully, responds, and uses questions to explore and stimulate students' thinking.
4. Enhance communication skills.

II. Scribe

1. Record points made by the group.
2. Help the group order their thoughts.
3. Participates in discussion.
4. Record resources used by the group.

III. Chair

1. Lead the group.
2. Encourage the group.
3. Maintain group dynamic.

IV. Timekeeper

1. Ensure the group keeps the task in hand.
2. Ensure scribe role.

V. Group members

1. Explore what they know and ask questions.
2. Tracking reports.
3. Participate in discussion.
4. Research all the objectives.
5. Share information.

❖ Advantages of PBL

1. **Student-centred:** fosters active learning, improved understanding, and retention and development of lifelong learning skills
2. **Generic competencies:** allow students to develop generic skills and attitudes desirable in their future practice
3. **Motivation:** It is fun for students and tutors, and the process requires all students to be engaged in learning.
4. **"Deep" learning:** fosters deep learning (students interact with learning materials, relate concepts to everyday activities, and improve their understanding)
5. **Constructivist approach:** Students activate prior knowledge and build on existing conceptual knowledge frameworks

❖ Disadvantages of PBL

1. **Tutors who can't "teach"**—Tutors enjoy passing on their own knowledge and understanding so may find PBL facilitation difficult and frustrating.
2. **Human resources:** More staff have to take part in the tutoring process
3. **Other resources**: Large numbers of students need access to the same library and computer resources simultaneously
4. **Lack of teacher role models**: Students may be deprived of access to a particular inspirational the teacher who in a traditional curriculum would deliver lectures to a large group.
5. **Information overload**—Students may be unsure how much self-directed study to do and what information is relevant and useful

Team application tAPP

tAPP (1)

Assurance of group members' participation and achievement of PBL objectives is a responsibility of:

a) Tutor.

b) Chair.

c) Scribe.

d) Group members.

tAPP (2)

A teacher decided to conduct a PBL session. He gives students a case for discussion and learning. Students are divided into small groups and identify and define the terms before writing their reading objectives.

Which step of PBL includes brainstorming, where students use their prior knowledge to identify the problem?

a) Step 1.

b) Step 2.

c) Step 3.

d) Step 4.

tAPP (3)

Dr Mohammed is a teacher at Ibn Sina University. he is trying to introduce a student-centered teaching method to the cardiovascular course, where students set in small groups discussing cardiovascular case scenarios to identify their reading knowledge and objectives and solve the case.

What is the learning method used by the teacher?

a) Lecture.

b) Problem-Based Learning.

c) Team Based Learning.

d) Community-Based Education.

Clarif the difficult terms/words
Step 1
Step 2
Define the problem
Brainstorming
Step 3
Step 4
Structuring and hypothesis
Formulating learning objectives
Step 5
Step 6
Self-directed learning
Sharing and evaluation
Step7

ANY Qs

Thank You For Your Attention

HAVE A NICE DAY....

Next:... Student feedback.